

Git & GitHub Documentation

What is Git?

Git is a tool that saves your work history. Think of it like a Time Machine for your files. Every time you save (called a 'commit'), Git remembers what your files looked like at that moment. If something breaks, you can go back to an earlier version.

What is GitHub?

GitHub is a website (github.com) where you store your Git projects online. Think of Git as your local save system and GitHub as Google Drive — it keeps your work safe online and lets others see or work on it too.

Installing & First-Time Setup

Before you start, you need to tell Git your name and email. This is only done once.

Command	What It Does (Simple Words)	Example
git config --global user.name "Your Name"	Tell Git your name (one-time setup)	git config --global user.name "Ravi Kumar"
git config --global user.email "you@email.com"	Tell Git your email (one-time setup)	git config --global user.email "ravi@gmail.com"
git config --list	Check your saved settings	git config --list
git --version	Check if Git is installed and see its version	git --version

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REM  SECTION 1: First-Time Setup – Tell Git who you are

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git config --global user.name "Next webby"

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git config --global user.email "nextwebby@gmail.com"

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git config --list
user.name=Next Webby
user.email=nextwebby@gmail.com
core.autocrlf=true
core.fscache=true

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git --version
git version 2.43.0.windows.1

Git is installed and configured successfully!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>
```

Section 1: Starting a Project

When you start using Git on a project folder, you are creating a 'repository' (short: repo). A repository is just a folder where Git is watching and tracking all changes.

Command	What It Does (Simple Words)	Example
git init	Start tracking a folder with Git (creates a new repo)	git init
git clone <URL>	Download a project from GitHub to your computer	git clone https://github.com/user/project
git status	See what files changed (use this all the time!)	git status

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REM  SECTION 2: Starting a Project – Initialize Git in your folder

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git init
Initialized empty Git repository in C:/Users/Next webby 01/Desktop/Daily Github works/New folder/.git/

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git status
On branch main

No commits yet

nothing to commit (create/copy files and use 'git add' to track)

Git is now watching this folder!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>
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REM  SECTION 2b: Download (Clone) a project from GitHub

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git clone https://github.com/nextwebby01/myproject
Cloning into 'myproject'...
remote: Enumerating objects: 45, done.
remote: Counting objects: 100% (45/45), done.
remote: Compressing objects: 100% (30/30), done.
Receiving objects: 100% (45/45), 12.34 KiB | 2.00 MiB/s, done.
Resolving deltas: 100% (8/8), done.

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>cd myproject

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git status
On branch main
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean

Project downloaded successfully!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>
```

Section 2: Saving Your Work

This is the most important part. Every time you make changes to your files, you follow these 3 steps:
Stage → Commit → Push.

Step 1: Stage	Step 2: Commit	Step 3: Push
Pick which files to save	Take a snapshot with a message	Upload to GitHub online

Command	What It Does (Simple Words)	Example
git add <filename>	Stage (select) one specific file to save	git add index.html
git add .	Stage ALL changed files at once	git add .
git commit -m "message"	Save a snapshot with a short description	git commit -m "Added login page"
git push	Upload your saved work to GitHub	git push
git push origin main	Upload to the 'main' branch on GitHub	git push origin main

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REM  SECTION 3: Saving Your Work – Stage then Commit

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.html
        style.css

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git add .

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git status
Changes to be committed:
        new file:   index.html
        new file:   style.css

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git commit -m "Added homepage and
[main (root-commit) alb2c3d] Added homepage and stylesheet
 2 files changed, 45 insertions(+)
 create mode 100644 index.html
 create mode 100644 style.css

Snapshot saved successfully!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>
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REM  SECTION 4: Push to GitHub – Upload your work online

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 1.23 KiB | 1.23 MiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/nextwebby/myproject.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.

REM  Next time, just use: git push
C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git push
Everything up-to-date

Code is now live on GitHub!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>
```

Section 3: Getting Updates from GitHub

When you are working with a team, others may upload changes. You need to download their changes to stay up to date.

Command	What It Does (Simple Words)	Example
git pull	Download and apply latest changes from GitHub	git pull
git fetch	Download changes but do NOT apply them yet	git fetch
git pull origin main	Pull changes from the 'main' branch specifically	git pull origin main

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REM SECTION 5: Pull from GitHub – Get latest changes from team

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
Updating a1b2c3d..e4f5g6h
Fast-forward
 index.html | 3 +++
 app.js     | 5 +++++
 2 files changed, 8 insertions(+)

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git pull origin main
From https://github.com/nextwebby/myproject
 * branch      main      -> FETCH_HEAD
Already up to date.

You now have the latest version!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>
```

Section 4: Branches

A branch is like a copy of your project where you can experiment freely. When your work is ready, you merge it back into the main project. This way the main project is never broken.

Example: The 'main' branch is the live website. You create a 'new-feature' branch to add a new button. When the button works perfectly, you merge it into main.

Command	What It Does (Simple Words)	Example
git branch	See all branches in your project	git branch
git branch <name>	Create a new branch	git branch new-feature
git checkout <name>	Switch to a different branch	git checkout new-feature
git checkout -b <name>	Create AND switch to a new branch (shortcut)	git checkout -b login-page
git switch <name>	Switch branch (newer, simpler command)	git switch main
git switch -c <name>	Create and switch to new branch (new style)	git switch -c my-feature
git merge <name>	Merge another branch into the current one	git merge new-feature
git branch -d <name>	Delete a branch (after merging)	git branch -d new-feature
git push -u origin <name>	Upload new branch to GitHub	git push -u origin new-feature

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REM SECTION 6: Branches – Work without breaking the main project

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git branch

* main

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git checkout -b new-login-page
Switched to a new branch 'new-login-page'

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git branch

main

* new-login-page

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git add . && git commit -m "Added
[new-login-page b7c8d9e] Added login page
1 file changed, 23 insertions(+)

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git push -u origin new-login-page
* [new branch] new-login-page -> new-login-page

Branch created and pushed to GitHub!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>

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REM  SECTION 6b: Merging – Bring your branch back into main

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git merge new-login-page
Updating a1b2c3d..b7c8d9e
Fast-forward
 login.html | 23 ++++++
 1 file changed, 23 insertions(+)

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git branch -d new-login-page
Deleted branch new-login-page (was b7c8d9e).

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git push
Writing objects: 100% (3/3), 890 bytes | 890.00 KiB/s, done.
To https://github.com/nextwebby/myproject.git
 a1b2c3d..b7c8d9e  main -> main

Branch merged and pushed successfully!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>
```

Section 5: Viewing History & Logs

You can see the full history of all saves (commits) in your project — who saved what and when.

Command	What It Does (Simple Words)	Example
git log	See full history of all commits	git log
git log --online	See a short, clean summary of commits	git log --online
git log --online --graph	See history as a visual tree/graph	git log --online --graph
git diff	See exactly what lines changed (not yet staged)	git diff
git diff --staged	See what lines changed in staged files	git diff --staged
git show <commit-id>	See details of one specific commit	git show a1b2c3d

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REM  SECTION 7: Viewing History – See all your saved commits

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git log --online
b7c8d9e (HEAD -> main) Added login page
a1b2c3d Added homepage and stylesheet
9f8e7d6 First commit - project setup

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git log --online --graph
* b7c8d9e (HEAD -> main, origin/main) Added login page
* a1b2c3d Added homepage and stylesheet
* 9f8e7d6 First commit - project setup

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git diff
diff --git a/index.html b/index.html
--- a/index.html
+++ b/index.html
+ <h1>Welcome to My App</h1>

Full history visible and trackable!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>
```

Section 6: Undoing Mistakes

Everyone makes mistakes! Git lets you undo changes safely.

Command	What It Does (Simple Words)	Example
git restore <file>	Undo changes in a file (go back to last commit)	git restore index.html
git restore .	Undo ALL changes in all files	git restore .
git restore --staged <file>	Unstage a file (undo git add)	git restore --staged style.css
git revert <commit-id>	Undo a commit safely (creates new commit)	git revert a1b2c3d
git reset --soft HEAD~1	Undo last commit but keep your file changes	git reset --soft HEAD~1
git reset --hard HEAD~1	Completely delete the last commit and changes	git reset --hard HEAD~1

Command	What It Does (Simple Words)	Example
git stash	Temporarily save changes without committing	git stash
git stash pop	Bring back your temporarily saved changes	git stash pop

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REM  SECTION 8: Undoing Mistakes – Nothing is permanent in Git!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git restore index.html
(file restored to last committed state)

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git restore --staged style.css
(style.css removed from staging area)

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git stash
Saved working directory and index state WIP on main: b7c8d9e

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git stash pop
On branch main
Changes not staged for commit:
    modified:   index.html
Dropped refs/stash@{0} (abc1234)

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git revert b7c8d9e
[main fle2d3c] Revert 'Added login page'

Mistake undone safely – history preserved!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>

```

Section 7: Connecting to GitHub

These commands help you link your computer project to GitHub for the first time.

Command	What It Does (Simple Words)	Example
git remote add origin <URL>	Connect your local project to GitHub	git remote add origin https://github.com/you/project
git remote -v	Check which GitHub URL is connected	git remote -v
git remote remove origin	Remove the GitHub connection	git remote remove origin
git push -u origin main	First push — sets up tracking with GitHub	git push -u origin main

```

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REM SECTION 9: Connecting Local Project to GitHub

REM Step 1: Create repo on github.com first, then connect:
C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git remote add origin https://git

REM Step 2: Check the connection is saved:
C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git remote -v
origin https://github.com/nextwebby/myproject.git (fetch)
origin https://github.com/nextwebby/myproject.git (push)

REM Step 3: Push your code to GitHub for the first time:
C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git push -u origin main
Enumerating objects: 4, done.
Writing objects: 100% (4/4), 1.23 KiB | 1.23 MiB/s, done.
To https://github.com/nextwebby/myproject.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.

Local project is now connected to GitHub!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>

```

Section 8: Everyday Workflow

Here is the daily routine most developers follow:

#	Command	Why You Do This
1	git pull	Get the latest changes from your team

2	git checkout -b my-feature	Create your own branch to work in
3	... make your changes ...	Edit your files as needed
4	git status	Check what files you changed
5	git add .	Stage all your changed files
6	git commit -m "What I did"	Save a snapshot with a clear description
7	git push origin my-feature	Upload your branch to GitHub
8	Open Pull Request on GitHub	Ask teammates to review your work

```

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REM SECTION 10: Full Daily Workflow – All steps in order

REM 1. Start your day – get latest changes:
C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git pull
Already up to date.

REM 2. Create your own branch to work in:
C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git checkout -b fix-navbar-bug
Switched to a new branch 'fix-navbar-bug'

REM 3. After editing files, stage and commit:
C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git add .
C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git commit -m "Fixed navbar alignn
[fix-navbar-bug c3d4e5f] Fixed navbar alignment on mobile
1 file changed, 4 insertions(+), 2 deletions(-)

REM 4. Push your branch to GitHub:
C:\Users\Next webby 01\Desktop\Daily Github works\New folder>git push origin fix-navbar-bug
* [new branch]      fix-navbar-bug -> fix-navbar-bug

Now go to GitHub and open a Pull Request!

C:\Users\Next webby 01\Desktop\Daily Github works\New folder>

```

Section 9: Git Glossary

Term	What It Means	Example / Think Of It As
Repository (Repo)	A folder where Git tracks all your file changes	<i>Your project folder</i>
Commit	A saved snapshot of your files at one moment	<i>Like pressing Save + taking a photo</i>

Term	What It Means	Example / Think Of It As
Branch	A separate copy of your project to work freely	<i>new-feature, bugfix, main</i>
Merge	Combine two branches together	<i>Merge new-feature into main</i>
Push	Upload changes to GitHub	<i>Send to the internet</i>
Pull	Download changes from GitHub	<i>Get from the internet</i>
Clone	Download a full copy of a GitHub project	<i>Get a full project copy</i>
Stage / Index	Selecting which files to include in next save	<i>git add selects the files</i>
Origin	The name for your GitHub connection	<i>Shortcut for your GitHub URL</i>
Main / Master	The main, official branch of the project	<i>The live, stable version</i>
Pull Request (PR)	Asking teammates to review and accept changes	<i>Done on GitHub website</i>
Conflict	Two people edited same lines — must fix manually	<i>Git asks you to decide</i>